

Phospho Localization Scoring & FLR Control

Plan v2: an LGBM localization score + per-ID FLR, with site-determining-ion features

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0. One-paragraph summary

Train an **LGBM model** to discriminate truly- from falsely-localized peptidoforms, using features that span three tiers: the **difficulty prior** $C(c,n)$, the existing **competition evidence** (**iso_weight_fraction**), and a new **site-determining-ion (SDI) spectral evidence** term (“S” features). Calibrate the model’s score against the positional-isomer decoys, target-decoy style, to assign **each $C(c,n)>1$ ID its own localization FLR** (a localization q-value). $C(c,n)\leq 1$ IDs are unambiguous and carry no localization FLR. Per-C stratification remains available as a validation lens / simpler fallback, but the LGBM (with $C(c,n)$ as a feature) is the primary approach. This v2 folds in review comments: LGBM-first, per-ID FLR (no single global target), MBR IDs accepted under the same FLR standard, a detailed S-feature recipe (correctness over efficiency for now), and an accurate description of what the decoys currently are.

1. Goal & success criteria

Goal. Assign every $C>1$ peptidoform passing 1% ID-FDR a **localization FLR** (per-ID, like a q-value): the estimated probability that its site assignment is wrong, calibrated from the decoys. Users filter on it directly (typical phospho operating point ~5-10% FLR - localization is usually less stringent than the 1% ID-FDR). No single target is baked in; we *report the per-ID value*.

Success criteria.

1. The LGBM localization score, target-decoy-calibrated, gives a monotone **per-ID localization FLR** that, at any cutoff, matches the realized decoy-FLR.
2. At a fixed FLR the LGBM recovers **more correctly-localized IDs than the raw frac** (and than per-C **frac** thresholds), i.e. the added features earn their keep — shown by an **ablation** (prior-only vs **frac** vs **frac+C** vs **frac+C+S**).
3. The decoy-FLR is **validated against the standards’ empirical FLR** (so far decoy runs conservative: ~9.6% decoy vs ~4% empirical at **frac** ≥ 0.75).

2. Foundation (what this rests on)

- ~91% of phospho IDs are $C>1$ (need a localization FLR); ~9% are $C=1$ (unambiguous, exempt).

- **frac** (`iso_weight_fraction_at_scan`) is a good confidence for **co-eluting** peptidoforms but is information-free for **singletons** (`iso_group_size==1` -> `frac=1` trivially).
- $C(c,n)$ = product over mod types of $C(\text{sites}, \text{present})$ is a strong monotone difficulty prior (singleton FLR by C: C1 3.2%, C2 6.6%, C3 11.2%, C4-6 20.3%, C7+ 38.0%) and rescues the singleton regime **frac** can't see.
- Localization already happens implicitly at two pipeline stages (fragment-index candidate selection, deconvolution ridge), both keyed on site-determining ions. The S features make that evidence explicit and usable by a model.
- Metric discipline: **per-file localization instances** (never max-frac dedup across files -> spurious 27%); **MBR accepted but held to the same FLR standard** (its decoys are transferred too, so the D/R ratio already prices MBR's effect in).

3. The model: LGBM over three feature tiers

One row per (peptidoform, file) localization instance with $C > 1$. Label for training = target vs positional-isomer decoy. Features:

(P) Difficulty prior - library-only, always defined. $C(c,n)$, $\log C$, per-type candidate-site count c , present-count n , min/mean distance between candidate sites, peptide length, whether the assigned site is terminal (b1/y1/y2 excluded from the library -> unlocalizable-ish).

(W) Competition evidence - from the existing search, free. `iso_weight_fraction_at_scan`, `iso_group_size_at_scan`, `weight_rank_at_scan`, `weight`, `mbr` flag.

(S) Site-determining-ion evidence - NEW (Section 4). S_C (min competitive ratio over alternatives), S_{mean} , $n_{\text{sdi_obs}}$ (observed site-determining ions), `sdi_support_frac`, and the **library-predicted** SDI intensity fraction (a prior for “is this even localizable” - the sparsity idea, computable without the observed spectrum).

LGBM handles the interactions (e.g. trust **frac** less when C is high; lean on S_C when `iso_group_size==1`). Per- C stratified thresholds stay as a simpler alternative and as a validation cross-check (does the model's FLR hold within each stratum?).

4. The S (site-determining-ion) features - detailed recipe

Correctness-first; optimize later. Computed per (peptidoform, file) at the instance's best-PSM scan (`scan_idx`). Reuses `site_determining.jl`, `matchVarMods/acceptor_positions`, and the SAME `permute_fragment_mz` the decoy generator uses.

4.1 Inputs

- Peptide `seq` (length L), and the assigned modification positions from `structural_mods` (per mod type; for a decoy, one assigned position is the shadow site).
- Candidate acceptor sites for the mod type: `acceptor_positions(matchVarMods(seq, var_mods))`. Define the comparison candidate set $\text{Cand} = (\text{real acceptor sites}) \cup (\text{assigned sites of this peptidoform})$. For a real peptidoform $\text{Cand} = \text{real acceptors}$; for a decoy it also includes the shadow site.
- The peptidoform's **library fragments**: for each fragment, (`base_type` in $\{b,y\}$, `frag_index` i , `frag_charge` z , `mz`, `predicted_intensity`) (from `detailed_fragments`

via `precursor_to_fragment_indices`).

- The **observed MS2 spectrum** at `scan_idx`: `peak (mz, intensity)` arrays from the run's Arrow file.
- Fragment mass tolerance (the search's ppm) for matching.
- `mod_mass` for the mod type (e.g. phospho 79.966331).

4.2 Which alternatives to compare

We use **single-move alternatives** (matching how decoys are built): for each assigned site `a` of the mod type and each *free* alternative site `a'` in `Cand \ {assigned}`, compare “mod on `a`” vs “mod on `a'`”. `S_C` is the worst (min) over all such pairs - the assigned placement must beat every alternative. (Multi-mod peptidoforms: iterate the assigned mods; the full `C(c,n)`-way joint enumeration is a later refinement - single-move covers the dominant ambiguity and is what the decoys model.)

4.3 Identify the site-determining ions for a pair (`a`, `a'`)

An ion is **site-determining** for (`a`, `a'`) iff its cleavage falls between them (it contains exactly one of `a`, `a'`), so the mod (+`mod_mass`) shifts its `m/z` under one placement but not the other. With `lo=min(a,a')`, `hi=max(a,a')`:

- **b ions**: `b_i` contains residues `1..i`; site-determining iff `lo <= i < hi` (i.e. `i` in `[lo, hi-1]`).
- **y ions**: `y_j` contains residues `L-j+1..L`; site-determining iff `lo <= L-j+1 <= hi` - equivalently the complementary cleavages, `j` in `[L-hi+1, L-lo]`.

There are $\sim 2 \cdot (hi-lo)$ such ions (`a` b and `a` y per interior cleavage), across each available fragment charge `z`. **Exclude neutral-loss channels** (phospho -98 H3PO4 and NL of mod-bearing fragments): they are not site-determining and can be isobaric with unmodified ions - including them makes decoys look artificially supported.

4.4 `m/z` under each placement, and matching

For each site-determining ion (given `base_type`, `i`, `z`):

- `m/z` under the **assigned** placement `a` is the library-predicted `mz` (the peptidoform is built with the mod on `a`).
- `m/z` under the **alternative** placement `a'` is `permute_fragment_mz(mz, base_type, i, z, L, a, a', mod_mass)` - the exact function the decoy generator uses to move a mod's contribution between sites.
- Match each `m/z` to the observed peaks within tolerance; take the matched peak intensity (0 if no match). Call them `I_a(ion)` and `I_{a'}(ion)`.

4.5 Support and the competitive ratio

`support(a | a') = sum over site-determining ions of I_a(ion)`

`support(a' | a) = sum over site-determining ions of I_{a'}(ion)`

`s(a, a') = support(a|a') / ( support(a|a') + support(a'|a) )` # undefined if `denom == 0`

`s` -> 1 when the observed spectrum's discriminating ions support the assigned site `a`; -> 0 when they favor the alternative `a'`; ~ 0.5 when ambiguous. **Undefined (`n_sdi_obs == 0` for this**

pair): unlocalizable against a' from this spectrum - flag, don't silently score.

4.6 Aggregate to per-instance features

S_C = min over alternatives a' of s(a, a') # must beat every alternative
S_mean = mean of s(a, a') over alternatives
n_sdi_obs = # distinct site-determining ions matched in the observed spectrum (any pair)
sdi_support_frac = (sum I_a over site-determining ions) / (sum intensity of ALL matched frags)
sdi_pred_frac = (sum predicted intensity of site-determining ions) / (sum predicted intensity of ALL matched frags)

sdi_pred_frac is the **library-only “localizability” prior** (Section 8, sparsity): if the ions that *would* distinguish the sites are predicted to be near-zero intensity, the peptidofrom is inherently hard regardless of the observed spectrum.

4.7 Decoys use the identical recipe

A loc-decoy's assigned site *is* its shadow site; its alternatives are the real acceptors. Its library fragments already carry the permuted (shadow-placement) m/z. Because the observed spectrum reflects the *real* peptide, the decoy's site-determining ions (supporting the shadow placement) should fail to match -> low S_C. That gap is exactly what calibrates the FLR. Computing S identically for reals and decoys is what makes the target-decoy FLR valid.

5. Per-ID localization FLR (calibration & reporting)

- Score every C>1 instance (real and decoy) with the LGBM.
- **Target-decoy q-value:** sort by score desc; at a score s , $FLR(s) = D(>=s) / R(>=s)$ (decoy vs real instance counts, C>1 only). Each real ID's **localization FLR = min FLR over cutoffs at or below its score** (monotone, q-value style). Report it as a per-ID column.
- **MBR IDs are included** in both R and (their transferred decoys) D, so they are held to the same standard automatically - no special-casing.
- **Report with-MBR and MBR-excluded** columns (MBR ~ +2 pts, informative).
- C==1 IDs: localization FLR = N/A (unambiguous); they still carry the ID-level 1% FDR.
- Cross-check the per-ID FLR against the standards' empirical FLR at matched cutoffs.

6. What the decoys currently ARE (accurate description) + realism gaps

Per peptide (grouping its peptidofoms):

1. `real_by = real_sites_by_mod(seq, var_mods)` - the real acceptor positions per mod type.
2. `shadow = shadow_decoy_sites(seq, real_by)` - draws, **per mod type, k = (number of real acceptor sites) positions UNIFORMLY AT RANDOM** from all **NON-acceptor residues** (`Random.shuffle(avail)[1:k]`, deterministic seed). **NOT biased toward the real sites** - they are random non-acceptor positions anywhere in the peptide. (The “short-distance-biased” draw is a *proposed* refinement, not what runs today.)
3. For each modified peptidofom with genuine ambiguity ($C(c,n)>1$ for some mod type), build ONE decoy: move one randomly-chosen mod of an ambiguous type from its real site to a random shadow site of that type; other mods stay put.

- Decoy fragments = parent fragments with the moved site’s m/z permuted (`permute_fragment_mz`); same sequence, **same precursor m/z** (mass preserved), `is_loc_decoy=true`, fresh `pair_id`.
- Decoy predicted iRT == parent iRT, exactly** (copied - Koina can’t predict a mod on a non-acceptor shadow residue).

Realism gaps (both make decoys potentially TOO EASY -> conservative/over-estimated FLR):

- Shadow sites are non-acceptor residues.** A real mislocalization lands on another *acceptor* (S/T/Y) site; a phospho on a random non-STY residue is easier for the search to reject. So the decoy tests a slightly different (easier) hypothesis than a true site-swap.
- Decoy iRT == parent iRT exactly** (confirmed unrealistic). Real positional isomers shift RT; ours co-elute perfectly with the parent, which (a) is unphysical and (b) makes MBR over-transfer decoys (recovered 29% vs 10% for reals). **Perturbing the decoy iRT** is the concrete fix.

These are Phase-4 levers (Section 9): pursue if the FLR calibration is off, but they don’t block Phases 1-3.

7. Validation & ablation

- Ablation** (headline): correctly-localized IDs at matched FLR for prior-only, `frac`, `frac+C`, `frac+C+S`. Quantifies each tier’s lift and whether S is worth the extraction.
- Per-C-stratum FLR calibration (does the single LGBM FLR hold within strata?).
- Standards ground-truth cross-check at several cutoffs.
- Per-dilution-point stability (decoy-FLR was flat ~7-8% across 256x).
- Coverage-vs-FLR curve vs baselines (global `frac`, winner-take-one).

8. Deferred / future (remembered, not now)

- SDI sparsity as an intrinsic limit.** Some precursors are just hard: if the site-determining ions have very low predicted relative intensity in the library spectrum, no method can localize them well. Captured now as the feature `sdi_pred_frac`; a later focus is a principled “unlocalizable” call (report peptide, not site) based on library predicted intensity.
- Multi-mod joint alternatives** (full `C(c,n)` enumeration) beyond single-move.
- Decoy realism fixes** (short-distance-biased shadow sites; decoy iRT perturbation).

9. Phased plan

Phase	Deliverable	Cost	Gate
1	S-feature extractor + per-instance feature table (existing results; reals + decoys)	analysis-only	do S features separate reals from decoys in high-C/singleton strata?
2	LGBM training + target-decoy per-ID FLR + ablation	analysis-only	does <code>frac+C+S</code> beat <code>frac+C</code> at matched FLR?

Phase	Deliverable	Cost	Gate
3	Wire per-ID localization FLR into reporting (+ tests); stratified cross-check	code + tests	per-ID FLR holds within C strata + vs empirical?
4 (opt)	Decoy realism (RT perturb / short-distance shadow) if calibration is off	generator + rebuild	decoy-FLR tracks empirical more tightly?

10. Resolved from review + remaining choices

Resolved: LGBM primary with `C(c,n)` as a feature (was Option B); per-ID FLR reported (no single target; typical 5-10%); MBR accepted under the same FLR standard; decoy realism deferred to Phase 4; SDI sparsity remembered as `sdi_pred_frac` + future work.

Remaining for you:

1. Training scope - train the LGBM **within CV folds on this dataset**, or hold out standards entirely as an independent test (cleaner validation, fewer training rows)?
2. Score target - train to predict “decoy vs real” (proxy), or, where we have standards ground truth, also fit/validate against **true correct-vs-incorrect** labels?
3. Do we want the per-ID FLR as a **q-value** (monotone, min-over-cutoffs) or a raw **PEP-style** local error probability, or both?

Companion docs: `site_determining_ion_score.{md,pdf}` (SDI worked example), `FINDINGS.md` (empirical basis), `isomer_decoy_shadow_sites.md` (decoy construction).